

palmsource™

PalmSource Seminar Series
Build Tomorrow ▸▸▸ Today

Presented By:
Ezekiel
Sanborn de
Asis

palm™
POWERED

Becoming More Competitive with New Platform Technology

Looking Better

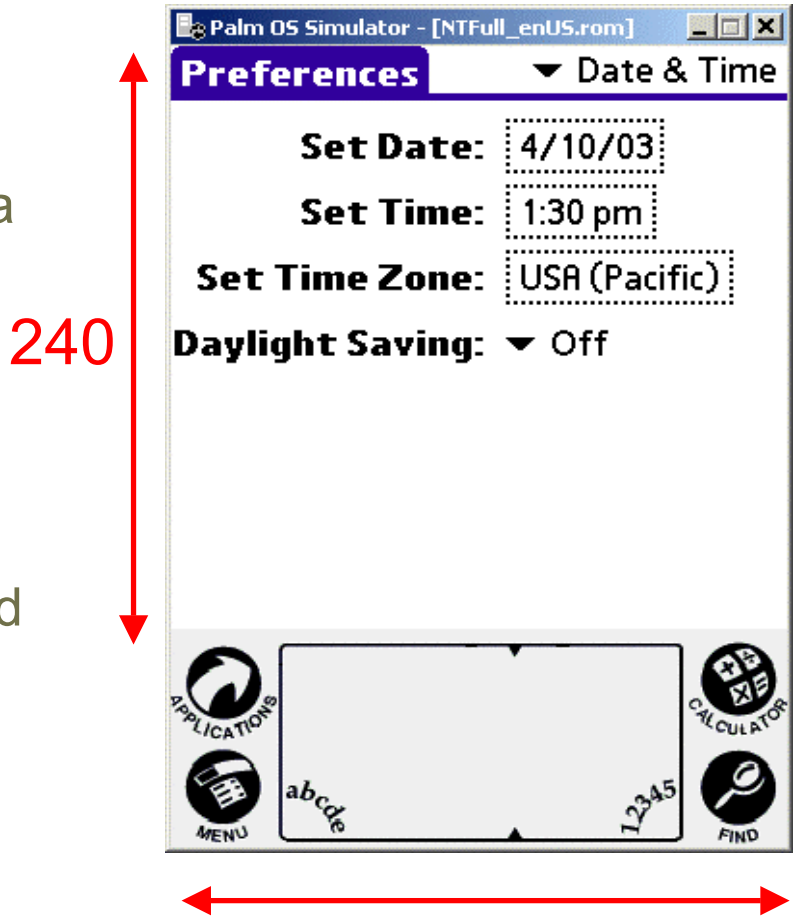
Sounding Better

Running Better

Doing More

Quarter VGA (QVGA)

- **1.5x density screen**
  - 240 x 320 Including input area
  - Active input with inking
  - Maybe collapsible
- **Much like 2x density**
  - bitmap and font families
  - 160 x 160 standard coord grid



QVGA Scaling

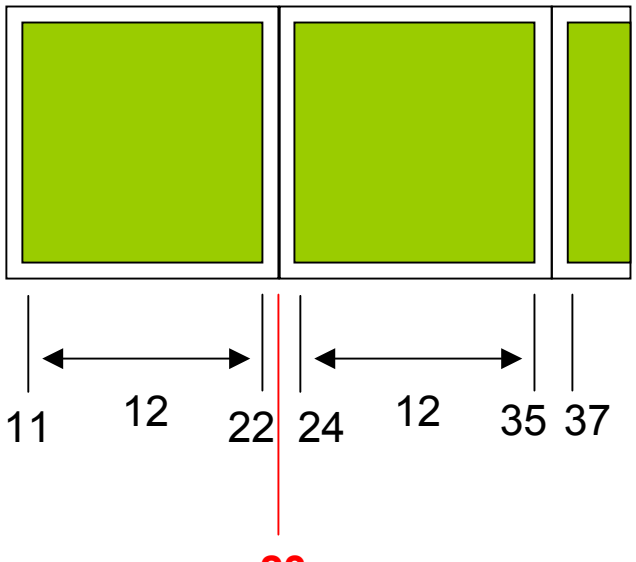
- **1.5 x odd number = rounding error**
- **Lines and pixels are rounded up**
- **Rectangle topLeft and extent are rounded down**
- **Every other pixel doubled for bitmaps**
  - Clipped to fit
- **No 2x ◂▸ 1.5x scaling**

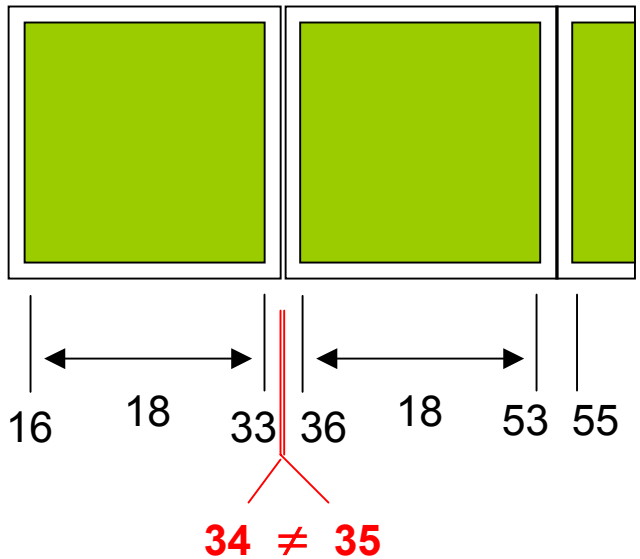
Form Layout for QVGA

- **Generally speaking use even values for topLeft coords**
- **Add 1.5x members to bitmap families**
- **Be extra careful when form objects are close together**
- **Use native coordinated for custom drawing**
- **Check knowledgebase for QVGA article**

Push Button Example

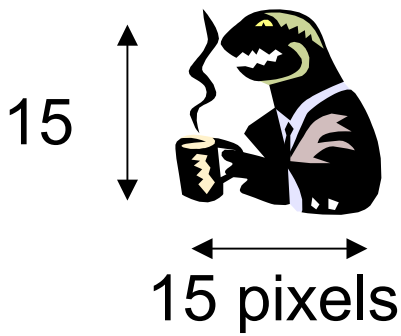
- **Rectangle frames are drawn around UI objects bounds**
  - Even extent values create odd boundaries





Offscreen Windows

- **Whenever possible use nativeFormat**
  - Other windows are low density
- **Be careful of rounding errors when offscreen window is sized based on the size of form or bitmap that it is buffering**



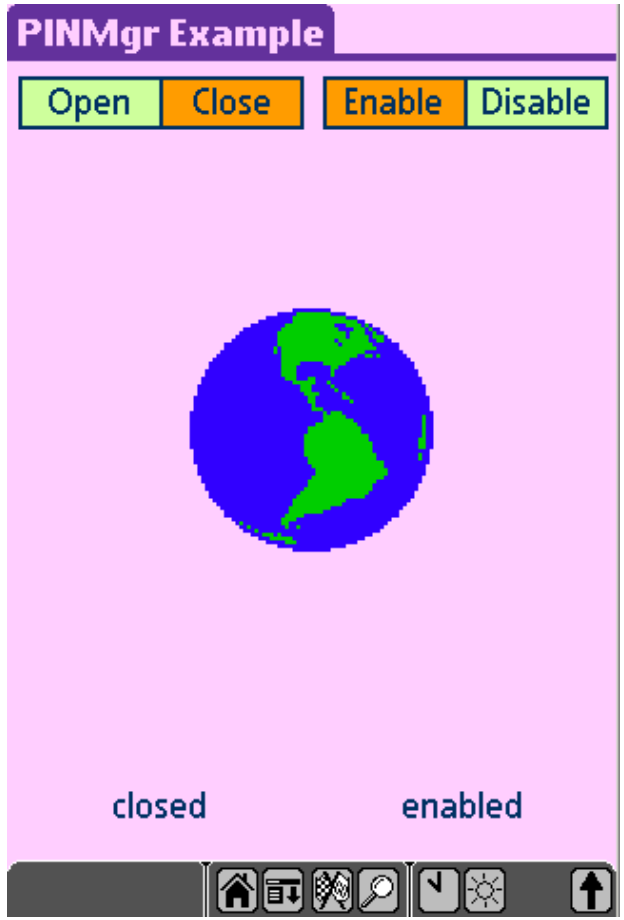
Offscreen
width =
3 x width of bitmap

Advanced "Bitmapping"

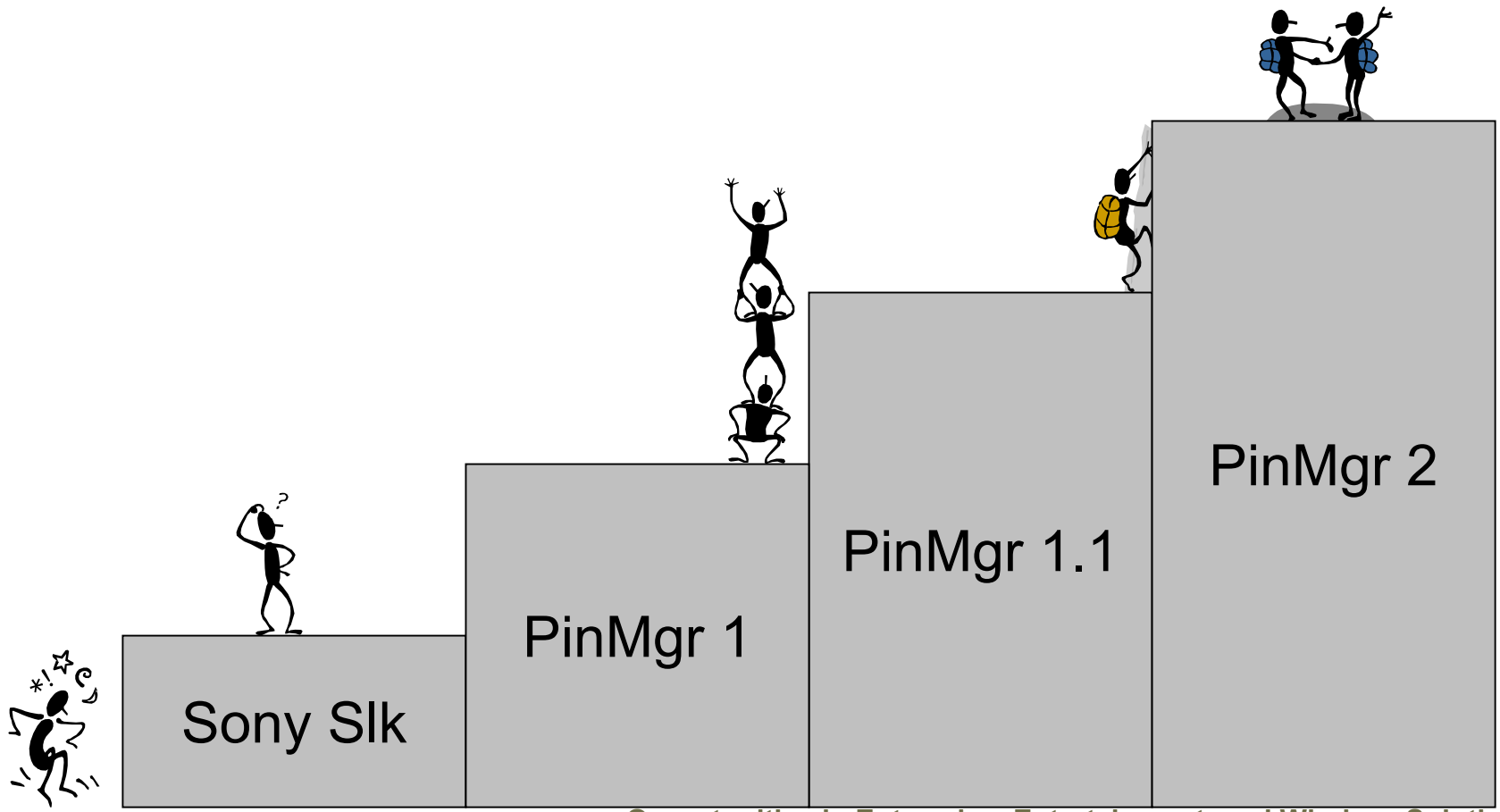
- **Common area for OS 5 issues**
  - See kb article 341 for FAQ and example code
- **General Tips**
  - Use bitmap families for most cases
  - See above mentioned example code for bitmaps created / manipulated at runtime
  - Use nativeFormat offscreen windows

Dynamic Input Area

- **2 New Features**
  - Inking
  - Collapsible
- **UX**
  - Hardware
- **Coming soon…**
  - Sample code for run anywhere app



Variations on a Theme



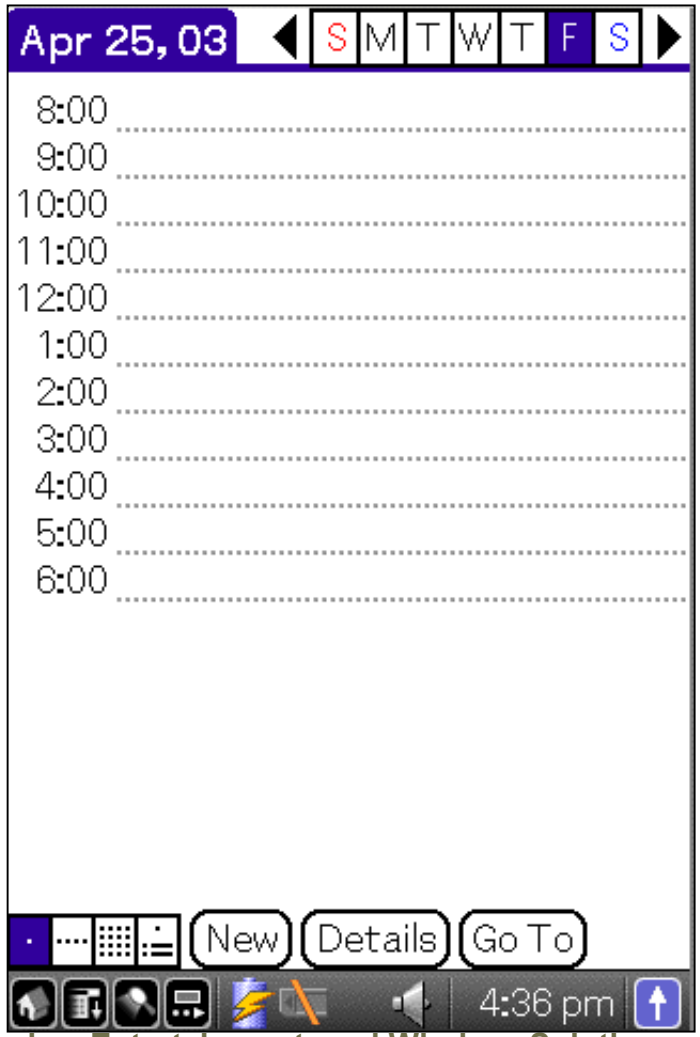
Sony Vsk Shared Lib

- **Documentation and headers in Sony SDK**
  - Version 1 and 2
- **Enable trigger**

`VskSetState`

- **Resize**

`sysNotifyDisplayChangeEvent`



PinMgr 1 and 1.1

- **Enable form by form**
  - frmLoadEvent or frmOpenEvent
- **PinMgr 1.1 adds additional functionality**
  - Pin Area user state
  - Show and Hide status bar area

Set up

- **In response to** `frmOpenEvent`

```
//This is the default behaivor of apps -- do not pull
up or push down input area
WinSetConstraintsSize(winH, 160, 160,
   pinMaxConstraintSize, …);
FrmSetDIAPolicyAttr(frmP, frmDIAPolicyCustom );
PINSetInputTriggerState(pinInputTriggerEnabled);

 PINSetInputAreaState(pinInputAreaUser); //PinMgr 1.1

//resize form move objects etc.
```

Reacting to a resize

- **In response to notification:**
  `sysNotifyDisplayResizedEvent`

Post Event

```
MemSet(&event, sizeof(EventType), 0);
event.eType = winDisplayChangedEvent;
EvtAddUniqueEventToQueue(&event, 0, true);
```

- **For PinMgr 1.0 also check for display windows extent changes after** `winEnterEvent`

Resize Form

```
WinGetDisplayExtent( &width, &height );
WinGetBounds( frmWinH, &frmWinRect );

offsetCoords = height - frmWinRect.extent.y -
   frmWinRect.topLeft.y;

if (isModal)
      offsetCoords -=2;

frmWinRect.extent.y += offsetCoords;
WinSetBounds( frmWinH, &frmWinRect );
```

Re-Layout the form

- **Moving, resizing and centering**
  - Buttons “stick” to bottom of screen
  - Gadgets, tables, lists “grow” and “shrink”
  - Some objects stick to other objects
  - Be careful when moving form bitmaps
- **Be very careful about object migration**

```
//move buttons to bottom
FrmGetObjectBounds(frmP, obIndex, &objectRect);
objectRect.topLeft.y += offsetCoords;
FrmSetObjectBounds(frmP, obIndex, &objectRect);
```

PinMgr 2

- **Pin input aware apps in Palm OS 6 will run by default "unaware"**
  - Even through PACE
- **Easy to add support and we give you support in advance**
  - Add "proper" constraints to your form
  - Handle different resize event

Looking Better

Sounding Better

Running Better

Doing More

OS 4 Sound Manager API

- **System creates wave on the fly**
  - Tungsten T does not clip sound to intended length properly resulting in sound lingering on

Checking for Sound

- **Check for Palm OS sound feature**
  - For Tungsten T Check device ID

```
FtrGet( sysFtrCreator, sysFtrNumOEMCompanyID, &Manufacturer );
FtrGet( sysFtrCreator, sysFtrNumOEMDeviceID, &DeviceID );
if (( Manufacturer == (UInt32)('Palm') && DeviceID == (UInt32)(
        'Frg1' )))
```

- **Sony Sound**
  - Yamaha ADPCM chip library from Sony

SndPlayResource

- **Tungsten T**
  - Problem with Async with certain sample rates
  - Fragmentation of heap
- **API does not exist on Sony**
- **Convenience API but limited**

Sound Streams

- **Use SndStreamCreate to create small number of streams on app init**
- **Blit PCM data onto stream buffer as needed**
  - Timing

Looking Better

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Doing More

Running Faster

- **System is ARM native and runs fast**
- **Look for small areas of heavy code**
- **Use ARMlets for optimization**
  - As per yesterday's talk

Running Slower

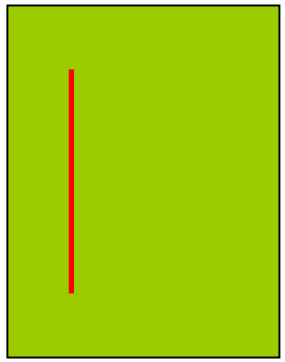
- **Many apps need to slow down to user perception / reaction time**
  - Graphics
  - Getting Pen Events
  - Scrolling

Running Cleaner

- **Shadow Structures is common cause for OS 5 incompatibilities**
- **Sound Manager Bug can cause fraged heap**
- **Graffiti 2 changes effects small number of apps**

Graffiti 2

- **Graffiti 2 Gotchas**





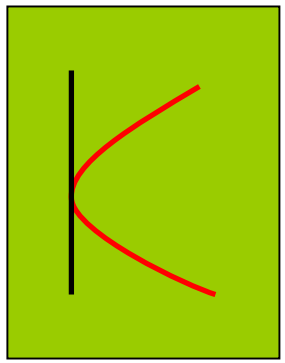
See article 397 in KB

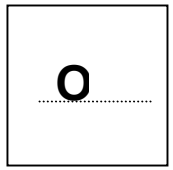
1) Sends 'L'

- Can not use L and K as menu shortcuts
- Apps process field text letter by letter need to be carefully

Graffiti 2

- Graffiti 2 Gotchas





See article 397 in KB

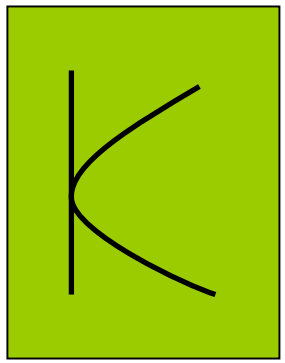
1) Sends 'L'

2) Sends backspace char

- Can not use L and K as menu shortcuts
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Graffiti 2

- Graffiti 2 Gotchas





See article 397 in KB

1) Sends 'L'

2) Sends backspace char

3) Sends 'K'

- Can not use L and K as menu shortcuts
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Looking Better

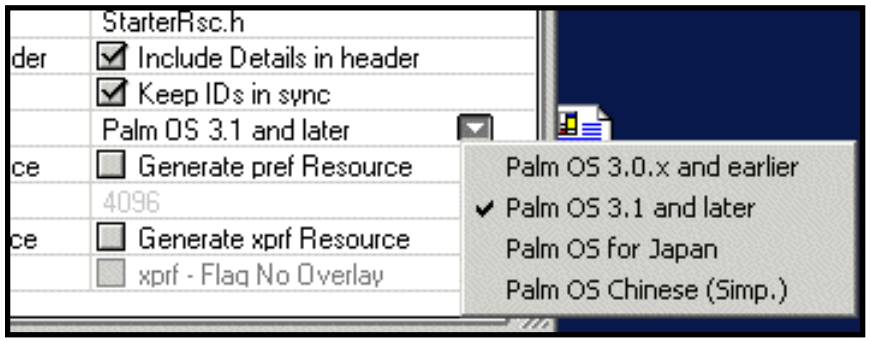
Sounding Better

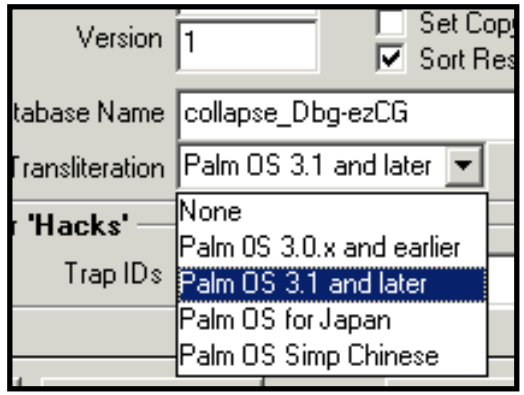
Running Better

Doing More

Chinese

- **Add Chinese resources with constructor**
  - MS Windows default language = Simp Chinese
- **Target Chinese with postlinker**
- **Works on official and non official GB char set based devices**





SDK

- **New Notifications:**
  - `sysNotifyInputAreaDrawingEvent`
  - `sysNotifyInputAreaPendownEvent`

  ****Do take to heart that some of these notifications go away in OS 6**
- **Additional Glue calls**

Navigation Buttons

- **Sony**
  - Jog dial SDK on Sony website
- **Palm, Inc.**
  - 5 way nav SDK in Dev Seeding Area
- **Garmin**
  - Header in Garmin SDK
- **General**
  - New Palm OS SDK defined generic navigation in chars.h and KeyMgr.h

Q&A